



Chasing the Neutrino Blazar Candidates. II. SED Modeling with a Hadronic Model

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Abstract

Blazars are promising candidates for high-energy neutrino sources, yet the physical origin of their neutrino emission remains uncertain. In this work, we extend our previous study by modeling the broadband spectral energy distributions (SEDs) of 103 neutrino blazar candidates (NBCs) within a hadronic framework. To estimate the maximum possible neutrino output, we adopt the assumption that the high-energy emission is dominated by $p\gamma$ interactions and the contribution from leptonic inverse Compton scattering is strongly suppressed. From the SED modeling, we constrain nine key parameters describing the emission region and particle energy distributions. We perform a partial-correlation analysis to investigate the relationship between neutrino luminosity and electromagnetic emission and find a weak-to-moderate correlation between the optical R band and neutrino emission. Our model predicts prominent proton synchrotron emission peaking in the MeV band for most sources, with 99 out of 103 NBCs exhibiting proton synchrotron peaks within 0.1–100 MeV, highlighting the MeV band as a key window for distinguishing between leptonic and hadronic scenarios. Based on the model-predicted maximum neutrino fluxes, we find that three NBCs are potentially detectable by IceCube, while up to 22, 45, and 62 sources may be detectable by KM3NeT, NEON, and TRIDENT, respectively. These results provide testable predictions for future multimessenger observations and offer new insights into the composition and radiation mechanisms of blazar jets.

Unified Astronomy Thesaurus concepts: Blazars (164); Jets (870); Neutrino astronomy (1100); High energy astrophysics (739)

Materials only available in the online version of record: figure set, machine-readable tables

1. Introduction

Blazars, the most extreme subclass of active galactic nuclei (AGNs), exhibit a range of remarkable observational properties, including rapid and strong variability, high and variable polarization, intense and variable γ -ray emission, and apparent superluminal motion (B. J. Wills et al. 1992; C. M. Urry & P. Padovani 1995; J.-H. Fan 2002; J.-H. Fan et al. 2004, 2014; B. Rani et al. 2013; M. Lyutikov & E. V. Kravchenko 2017; H. Xiao et al. 2019; H. B. Xiao et al. 2022). Based on their optical spectral characteristics, blazars are commonly divided into two subclasses: BL Lacertae objects (BL Lacs) and flat-spectrum radio quasars (FSRQs). BL Lacs are characterized by weak or absent emission lines, with rest-frame equivalent widths $EW < 5 \text{ \AA}$, whereas FSRQs exhibit prominent broad emission lines with $EW \geq 5 \text{ \AA}$ (C. M. Urry & P. Padovani 1995; R. Scarpa & R. Falomo 1997). Blazar emission is characterized by a two-bump broadband spectral energy distribution (SED) in the $\log \nu$ – $\log \nu F_\nu$ representation. The low-energy component, peaking from the millimeter to soft X-ray bands, is generally attributed to synchrotron radiation

from relativistic electrons. In contrast, the origin of the high-energy component, which typically peaks in the MeV–GeV range, remains debated and is commonly interpreted within either leptonic or hadronic emission scenarios.

The development of leptonic models has been well established over several decades. In the 1960s, it was recognized that the nonthermal emission from radio sources, covering the radio to optical bands, could be attributed to synchrotron radiation from relativistic electrons (K. C. Westfold 1959; V. L. Ginzburg & S. I. Syrovatskii 1965; K. I. Kellermann 1966). It was subsequently realized that these same electrons can produce high-energy photons through inverse Compton (IC) scattering off ambient soft photon fields, such as synchrotron photons (T. W. Jones et al. 1974; G. Ghisellini et al. 1985), photons from the broad-line region (M. Sikora et al. 1994; Z. Fan et al. 2006), or photons from the dusty torus (DT; M. Błażejowski et al. 2000; C. Arbeiter et al. 2002; A. Sokolov & A. P. Marscher 2005). The leptonic scenario was further consolidated following the detection of γ -ray emission by EGRET (L. Maraschi et al. 1992), and it has since become the most widely accepted framework for modeling blazar SEDs, particularly after the launch of the Fermi Large Area Telescope (Fermi-LAT). Given that cosmic rays are predominantly composed of protons, it was natural to consider whether protons might also play a significant role in blazar jets, thereby motivating the development of hadronic models for blazar emission. In early 1990s,



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K. Mannheim (1993) proposed that protons within the jet are accelerated to ultra-high energies and subsequently interact with ambient photon fields through photomeson processes ($p\gamma$ interactions) to produce secondary particles, which in turn emit γ -rays. F. A. Aharonian (2000) employed hadronic models to explain the TeV emission from BL Lac objects, although such models often require jet powers that exceed the Eddington limit (M. Böttcher et al. 2013). The detection of high-energy neutrino emission has provided the first multimessenger evidence for the presence of relativistic hadrons in blazar jets (IceCube Collaboration et al. 2018). Consequently, hadronic models have gained more attention because they produce neutrinos through charged-pion decay chains, and this mechanism has been naturally applied to explain the neutrino detection from TXS 0506+056 and to predict neutrino emission for other sources (M. Cerruti et al. 2019; R. Xue et al. 2019, 2021; Z.-R. Wang et al. 2024; Z. Ouyang et al. 2025).

Neutrino astronomy is widely regarded as a promising avenue for addressing the century-old questions surrounding the origin of cosmic rays and the mechanisms responsible for their acceleration. In the context of blazar studies, neutrinos serve as powerful probes of jet composition, the physical environment near supermassive black holes, and the contribution of blazars to the diffuse high-energy neutrino background. These fundamental questions can, in principle, be investigated through neutrino-emitting blazars; however, at present, TXS 0506+056 remains the only blazar with a firmly established neutrino association (IceCube Collaboration et al. 2018).

In our previous work (J. Zhu et al. 2024, hereafter Paper I), we employed a few-shot learning approach to identify neutrino blazar candidates (NBCs) from Fermi-LAT-detected blazars (S. Abdollahi et al. 2022), resulting in a sample of 199 NBCs. In this work, with the aim of extending our previous study and investigating the physical properties of NBCs, we adopt a hadronic model to construct broadband SEDs for these sources. Similar modeling frameworks have been adopted in previous studies (e.g., M. Cerruti et al. 2015; H. Abe et al. 2025), although several modifications are introduced in the present work. In order to estimate the maximum possible neutrino flux, we assume that all high-energy (hereafter HE) and very-high-energy (hereafter VHE) photons are produced through $p\gamma$ interactions and the associated electromagnetic cascades. This paper is arranged as follows. In Section 2, we introduce the SED-fitting method; in Section 3, we present the results; in Section 4, we present the discussions; and in Section 5, we summarize our conclusions.

2. Sample and Method

We searched for multiwavelength observations for the 199 NBCs (see Paper I) by matching the associated names provided in the Fermi-LAT Fourth Source Catalog (4FGL; S. Abdollahi et al. 2022) with entries in the Firmamento database (D. Tripathi et al. 2024).⁹ During the data collection process, we identified 103 NBCs with sufficient multiwavelength coverage that exhibit a clear, characteristic two-humped blazar SED structure by visual inspection. These data span from the radio to the γ -ray bands and were obtained from different observational campaigns. In addition, TXS 0506

+056 is included in this work, and its multiwavelength data are used to construct the SED and to compare its properties with those of the NBCs. Typically, two approaches can be adopted when handling multiwavelength data prior to broadband SED construction. One approach is to select quasi-simultaneous data within a narrow time window, which is suitable for constraining emission-region properties and testing specific SED models (e.g., C. Tan et al. 2020; Z.-R. Wang et al. 2024). Alternatively, unfiltered archival data can be used to investigate the general properties of a blazar population (e.g., H. Xiao et al. 2025). In this work, we adopt the latter approach and use the multiwavelength data without temporal filtering. Consequently, both quiescent and flaring states are included in the broadband SEDs constructed for the 103 sources, enabling us to investigate their general, population-level properties rather than state-dependent behaviors. Motivated by the current understanding that blazar jets are composed of both leptons and hadrons, we employ a hadronic model to construct the broadband SEDs in this study.

The one-zone hadronic model is employed to reproduce the broadband and neutrino emissions using the self-consistent, time-dependent numerical code AM³ (S. Gao et al. 2017; M. Klinger et al. 2024). We assume a spherical emission “blob” region of radius R_{em} , immersed in a homogeneous magnetic field of strength B , moving with a bulk Lorentz factor Γ along the relativistic jet. This results in a Doppler factor $\delta = \frac{1}{\Gamma(1 - \beta \cos \theta)}$, where βc is the velocity of the emission blob, c is the speed of light, and θ is the viewing angle of the jet with respect to our line of sight. Given that the viewing angle is typically less than a few degrees for blazars ($\theta \leq 1/\Gamma$), $\Gamma \simeq \delta$ can be obtained. We assume that nonthermal electrons and protons are accelerated and injected into the emission blob with an injection rate following a power-law spectrum $\frac{d^2 N_{e(p)}}{d\gamma_{e(p)} dt} \propto \gamma^{-\alpha_{e(p)}}$ ranging from $\gamma_{e(p)}^{\min}$ to $\gamma_{e(p)}^{\max}$, where $\gamma_{e(p)}$ and $\alpha_{e(p)}$ denote the Lorentz factor and spectral index of the electrons and protons, respectively.¹⁰ The injected power in electrons and protons is characterized by the respective injection luminosity $L_{e(p),\text{inj}}$. The time-dependent transport equation is then solved to obtain the steady-state particle distributions:

$$\frac{\partial N(\gamma, t)}{\partial t} = Q(\gamma, t) - \frac{\partial}{\partial \gamma}(\dot{\gamma}(\gamma, t)N(\gamma, t)) - \frac{N(\gamma, t)}{t_{\text{esc}}}, \quad (1)$$

where $Q(\gamma, t)$ denotes the particle injection rate, $\dot{\gamma}(\gamma, t)$ is the energy-loss rate, and t_{esc} is the particle escape timescale. The escape timescale is assumed to be comparable to the dynamical timescale, $t_{\text{esc}} \sim t_{\text{dyn}} = R/c$, which is widely adopted in the literature (e.g., R. Prince et al. 2024; X. Rodrigues et al. 2024; Z.-R. Wang et al. 2024; X. Jiang et al. 2025). A steady state is achieved when particle injection is balanced by cooling and/or escape within the emission region. In the steady-state, electrons produce synchrotron radiation and contribute to the low-energy hump after synchrotron self-absorption. Protons interact with

⁹ https://firmamento.nyuad.nyu.edu/data_access

¹⁰ Quantities with the superscript “obs” refer to the observed frame, while those with “rest” refer to the rest frame of the AGN; quantities without a superscript are defined in the rest frame of the relativistic jet.

Table 1
The Model Parameter Space

Parameter	Space	Units
$\log R_{\text{em}}$	(14, 17)	cm
$\log B$	(−2, 2)	G
Γ	(1, 30)	...
$\log L_e$	(38, 45)	erg s ^{−1}
$\log \gamma_e^{\text{min}}$	(1, 5)	...
$\log \gamma_e^{\text{max}}$	(3, 6)	...
α_e	(1, 3.5)	...
$\log L_p$	(42, 47)	erg s ^{−1}
$\log \gamma_p^{\text{min}}$	Fixed to 2.0	...
$\log \gamma_p^{\text{max}}$	(5, 9)	...
α_p	Fixed to 1.0	...
$\log \nu_{\text{thermal}}^{\text{rest}}$	(14, 15)	Hz
$\log L_{\text{thermal}}^{\text{rest}}$	(40, 48)	erg s ^{−1}

the photons in the synchrotron radiation field and contribute to the HE hump through photo-pion production ($p\gamma \rightarrow n\pi^+$ or $p\pi^0$) and Bethe–Heitler pair production ($p\gamma \rightarrow pe^+e^-$). Moreover, HE photons can undergo photon–photon pair production and annihilation with low-energy photons ($\gamma\gamma \rightarrow e^+e^-$). Neutrinos are produced through charged-pion π^+ decay ($\pi^+ \rightarrow \mu^+\nu_\mu$), followed by subsequent μ^+ decay ($\mu^+ \rightarrow e^+\nu_e\bar{\nu}_\mu$) from the photo-pion production. The secondary electron–positron pairs produced by these processes, including photo-pion production, Bethe–Heitler pair production, and photon–photon annihilation, can subsequently radiate via synchrotron and IC processes. To estimate the neutrino emission as an upper limit from purely hadronic origins, we assume that the HE hump is dominated by hadronic processes and suppress the leptonic IC scattering component. At very high energies, γ -ray photons interact with the extragalactic background light (EBL), resulting in significant attenuation of the observed flux. This effect is included in our modeling using the EBL model of A. Domínguez et al. (2011), applied after modeling for each source. In the present work, the low-energy component of the SED, spanning the radio to the optical/ultraviolet or X-ray bands, is attributed to synchrotron radiation from relativistic electrons. At higher energies, the X-ray emission is interpreted as synchrotron radiation from either relativistic electrons or protons, while the GeV γ -ray emission is attributed to hadronic interactions, predominantly via Bethe–Heitler pair production.

For some sources, we employ a blackbody thermal function to fit an additional bump appearing in the frequency range 10^{14} – 10^{15} Hz, which is characterized by its peak frequency $\nu_{\text{thermal}}^{\text{rest}}$ and thermal luminosity $L_{\text{thermal}}^{\text{rest}}$.

Based on the physical setup described above, the one-zone hadronic model is characterized by 11 parameters; for sources with an additional thermal hump, two more parameters are included, bringing the total to 13 parameters. The model parameter spaces for all parameters are listed in Table 1. In particular, we fix the minimum Lorentz factor of protons to $\gamma_p^{\text{min}} = 10^2$, and the proton spectral index to $\alpha_p = 1.0$. During the fitting process, radio data below 10^{11} Hz are excluded as the one-zone model cannot adequately reproduce the observed radio emission due to synchrotron self-absorption. We perform

SED fitting of the multiwavelength data using the PySwarms package,¹¹ which implements the particle swarm optimization (PSO) algorithm (L. J. V. Miranda 2018). This method simulates a population of particles that collectively explore the parameter space to minimize the objective function and is well-suited for exploring high-dimensional parameter spaces. We note, however, that the PSO method may not always converge to the global optimum, but can instead settle in a local minimum, potentially leading to suboptimal SED representations. To mitigate this effect, an additional “fit-by-eye” procedure is employed to refine the parameters and obtain an acceptable representation of the SED.

3. Results

3.1. Results of SED Fitting

Our sample consists of five FSRQs, 13 blazar candidates of uncertain type (BCUs), and 85 BL Lac objects. The sources span a redshift range from $z = 0.01$ to $z = 2.340547$. The broadband SEDs of all sources are presented in Figure 1, while the parameters describing the emission region, as well as the electron and proton energy distributions, are summarized in Table 2.

The distributions of the nine free parameters in our model are shown in Figure 2. We find that the magnetic field strength tends to approach the assumed upper limit, with 68 out of 103 sources (66.0%) exhibiting $\log B \geq 1.0$. This behavior arises naturally from our modeling strategy, in which the low-energy hump of the blazar SED is attributed to synchrotron radiation from relativistic electrons, with the electron energy assumed to be primarily dissipated through the synchrotron process. Consequently, the model requires relatively strong magnetic field strengths. To suppress the IC contribution and maximize the neutrino output, a relatively strong magnetic field is required to efficiently dissipate the electron energy through synchrotron emission, thereby allowing protons to dominate the γ -ray emission. The average energy ratio between protons and electrons is $\log(P_p/P_e) \simeq 3.73 \pm 0.78$, while that between protons and magnetic field is $\log(P_p/P_B) \simeq 1.67 \pm 1.03$. Therefore, the power carried by relativistic protons dominates over both the power carried by relativistic electrons and that carried by the magnetic field.

3.2. Correlation between Neutrino Emission and Electromagnetic Radiation

In this work, we collected the masses of the central supermassive black holes and the luminosities of the accretion disks from V. S. Paliya et al. (2021), radio flux densities, optical magnitudes, and X-ray flux from the BZCAT,¹² and γ -ray photon fluxes together with the corresponding photon indices from 4FGL. These data are listed in Table 3. The multiwavelength intensities are converted into luminosities through

$$L = 4\pi d_L^2 (1+z)^{\alpha-1} F, \quad (2)$$

where F is the flux in a given band, α is the corresponding spectral index, and $d_L = \frac{c}{H_0} \int_1^{1+z} \frac{1}{\sqrt{\Omega_m x^3 + 1 - \Omega_m}} dx$ is the

¹¹ <https://pyswarms.readthedocs.io/en/latest/index.html>

¹² <https://www.ssdc.asi.it/bzcat/>

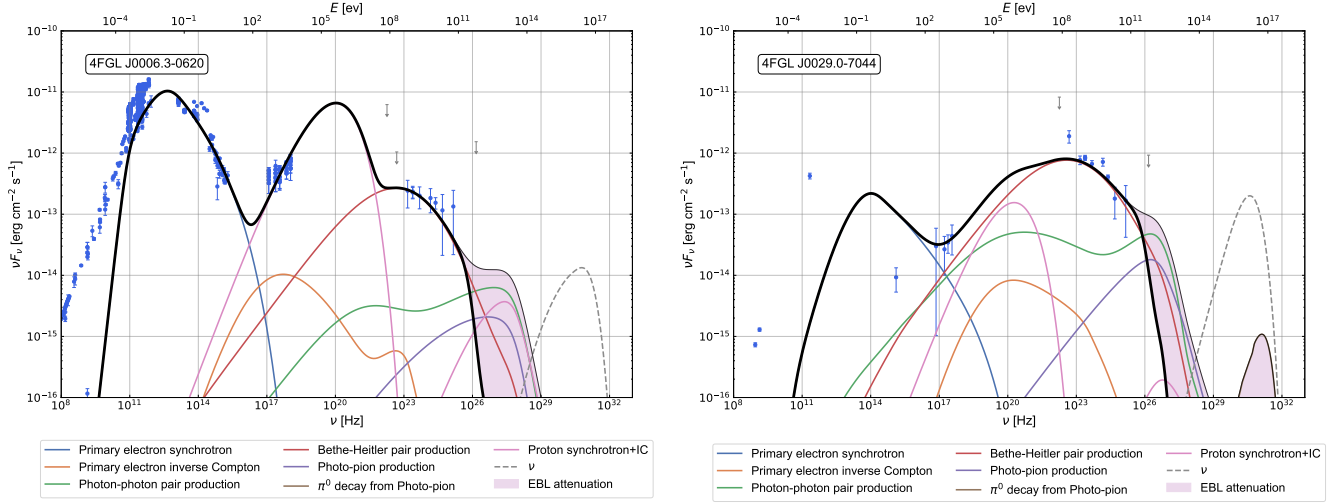


Figure 1. SED modeling plots for the 103 NBCs. The blue curve represents the electron synchrotron emission; the green curve represents emission from photon-photon pair production; the red curve represents emission from Bethe-Heitler pair production; the purple curve represents emission from photon-pion production; the brown curve represents proton synchrotron emission; the pink curve represents emission from π^0 decay; the dashed gray curve represents neutrino emission; the pink shaded region indicates attenuation by the extragalactic background light (EBL); and the solid black curve represents the summed model spectrum. Only the first two plots (4FGL J0006.3-0620 and 4FGL J0029.0-7044) of the NBCs are shown here; the entire version is displayed online.

(The complete figure set (103 images) is available in the [online article](#).)

Table 2
SED Modeling Parameters

4FGL Name	Class	z	$\log R_{\text{em}}$ (cm)	$\log B$ (G)	Γ	$\log L_e$ (erg s $^{-1}$)	$\log \gamma_e^{\text{min}}$	$\log \gamma_e^{\text{max}}$	α_e	$\log L_p$ (erg s $^{-1}$)	$\log \gamma_p^{\text{max}}$	νF_ν (erg cm $^{-2}$ s $^{-1}$)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
J0006.3-0620	BL Lac	0.346676	17.00	0.449	10.29	41.95	2.31	3.82	2.90	45.81	7.53	1.33E-14
J0029.0-7044	BLL	0.27	15.06	0.947	7.17	40.71	2.80	5.26	2.81	45.23	7.36	2.00E-13
J0045.7+1217	BLL	0.255	14.82	1.294	12.86	40.65	2.27	5.08	1.99	43.90	7.89	2.03E-12
J0121.8-3916	BLL	0.3	15.84	0.720	11.50	40.75	3.46	5.30	2.40	45.13	7.54	8.67E-13
J0123.1+3421	BLL	0.27	15.55	0.800	17.70	40.74	4.07	4.86	1.08	44.64	7.22	2.31E-12

Note. Column (1): 4FGL name. Column (2): classification. Column (3): redshift. Column (4): logarithmic dissipation region size, R_{em} (in units of centimeters). Column (5): magnetic field strength, B (in units of gauss). Column (6): bulk Lorentz factor. Column (7): luminosity of injected relativistic electrons. Column (8): minimum energy of electron population. Column (9): maximum energy of electron population. Column (10): spectral index of electron distribution. Column (11): luminosity of injected relativistic protons. Column (12): maximum energy of proton population. Column (13): flux of neutrino emission. Note, only five items are displayed.

(This table is available in its entirety in machine-readable form in the [online article](#).)

Table 3
The Central Engine and Multiwavelength Emission

4FGL Name	$\log M_\odot$ ($M_\odot$)	$\log L_d$ (erg s $^{-1}$)	$f_{1.4 \text{ GHz}}$ (mJy)	m_R (mag)	F_x (erg cm $^{-2}$ s $^{-1}$)	Flux1000 (ph cm $^{-2}$ s $^{-1}$)	α_{ph}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
J0006.3-0620	8.93	44.52	2051	17.9	0.82	1.21E-10	2.17
J0029.0-7044	...	...	94	16.7	...	3.93E-10	2.31
J0045.7+1217	8.82	44.18	104	17.4	0.56	9.85E-10	1.95
J0121.8-3916	8.58	44.39	...	...	...	2.50E-10	1.93
J0123.1+3421	...	...	44	17.9	25.3	2.30E-10	1.69

Note. Column (1): 4FGL name. Column (2): black hole mass (in units of solar mass). Column (3): accretion disk luminosity. Column (4): flux density at 1.4 GHz. Column (5): optical R -band magnitude. Column (6): X-ray flux in the range 0.1–2.4 keV. Column (7): γ -ray integral photon flux from 1 to 100 GeV. Column (8): γ -ray photon index of “Flux1000”. Note, only five items are displayed.

(This table is available in its entirety in machine-readable form in the [online article](#).)

luminosity distance (E. Komatsu et al. 2011). During the calculations, a radio spectral index of $\alpha_r = 0$ is adopted for all sources in our sample; an optical spectral index of $\alpha_o = 0.5$ is adopted for BL Lacs and $\alpha_o = 1$ for the remaining sources;

an X-ray spectral index of $\alpha_x = 1.3$ is adopted for BL Lacs, $\alpha_x = 0.78$ for FSRQs, and $\alpha_x = 1.05$ for BCUs; and the γ -ray spectral index is taken to be $\alpha_\gamma = \alpha_{\text{ph}} - 1$. These assumptions have also been employed in the literature

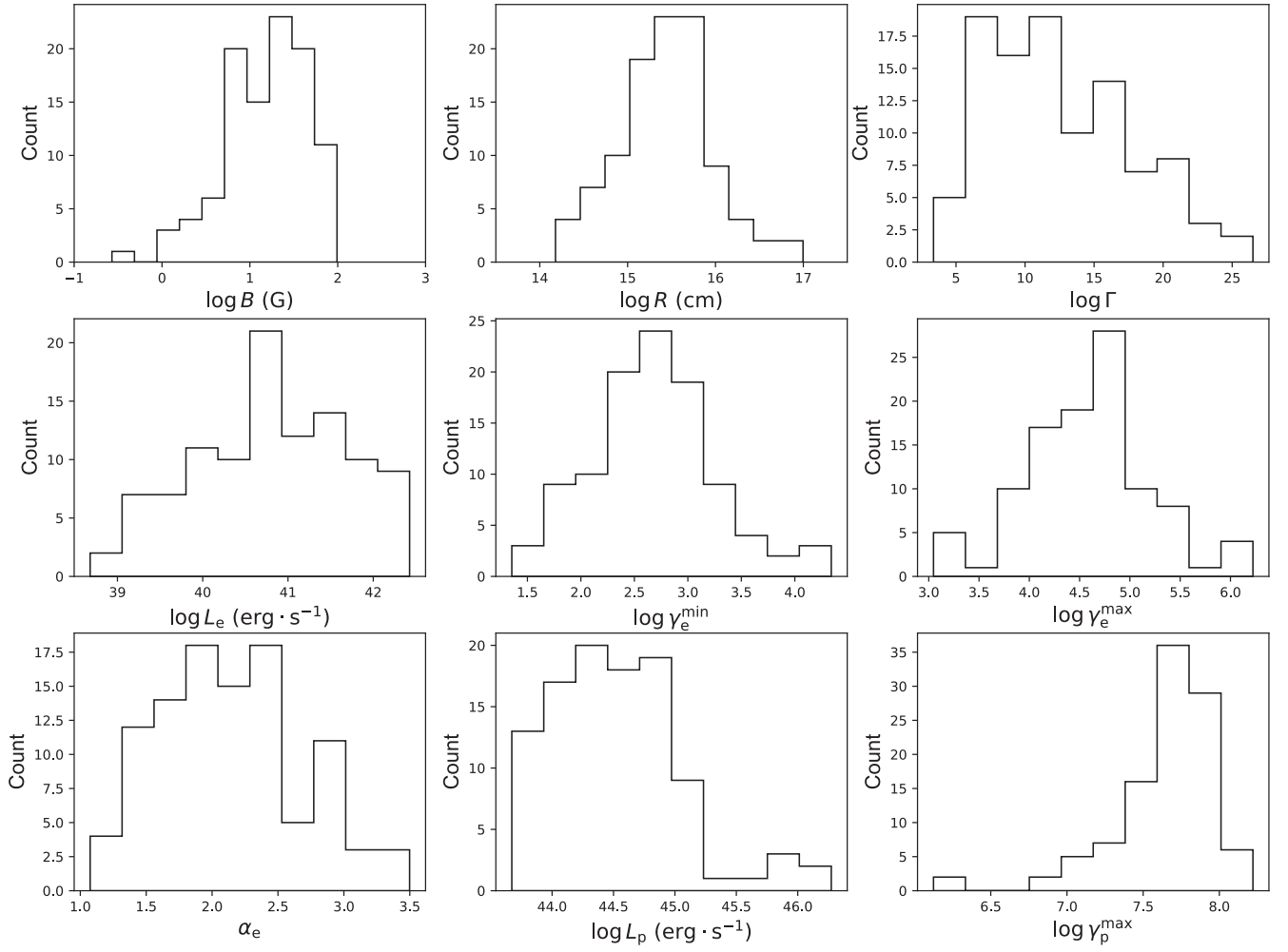


Figure 2. Distribution of SED modeling parameters

(e.g., D. Donato et al. 2001; A. A. Abdo et al. 2010; J. H. Fan et al. 2016). The optical magnitudes are corrected for foreground Galactic extinction using values collected from the NASA/IPAC Extragalactic Database.¹³ We explored the correlations between the neutrino luminosity ($\log L_\nu$) and the 1.4 GHz radio luminosity ($\log L_{1.4\text{GHz}}$), optical R -band luminosity ($\log L_R$), X-ray luminosity ($\log L_X$), and γ -ray luminosity ($\log L_\gamma$). The correlations were fit in the form $y = \beta x + a$ using ordinary least-squares (OLS) bisector regression. The results are illustrated in Figure 3 and presented in Table 4. Considering that luminosity is correlated with redshift, we removed the influence of redshift to derive the intrinsic correlations among the luminosities through a partial-correlation analysis:

$$r_{ij,k} = \frac{r_{ij} - r_{ik}r_{jk}}{\sqrt{(1 - r_{ik}^2)(1 - r_{jk}^2)}}, \quad (3)$$

in which r_{ij} , r_{ik} , and r_{jk} are the correlation coefficients between the variable pairs i and j , i and k , and j and k , respectively. The variables i and j represent luminosities, while the variable k represents redshift.

The partial Pearson correlation analysis indicates positive correlations between $\log L_\nu$ and $\log L_R$, $\log L_X$, and $\log L_\gamma$. We find that the strength of the correlation increases with the photon energy of the considered electromagnetic bands. Among the four bands examined, the correlation between $\log L_\nu$ and $\log L_\gamma$ is the strongest, followed by that between $\log L_\nu$ and $\log L_X$. Within the framework of the hadronic model, the HE hump of the blazar SED is primarily produced by proton synchrotron radiation and Bethe–Heitler pair production. Proton synchrotron emission mainly contributes to the correlation involving the X-ray band, while Bethe–Heitler processes dominate the γ -ray emission. Both mechanisms involve interactions of HE protons, and the resulting HE photons can act as seed photons for photo-pion production, ultimately leading to neutrino emission. There is weak-to-moderate correlation between $\log L_\nu$ and $\log L_R$, while no significant correlation is found for $\log L_\nu$ and $\log L_{1.4\text{GHz}}$.

3.3. Correlation between Neutrino Emission and the Central Engine

We investigate the potential connection between neutrino emission and the central engine properties, specifically the mass of the central supermassive black hole (M_{BH}) and the accretion disk luminosity (L_{disk}). These quantities are adopted from V. S. Paliya et al. (2021), where M_{BH} and L_{disk} are

¹³ <https://ned.ipac.caltech.edu/forms/byname.html>

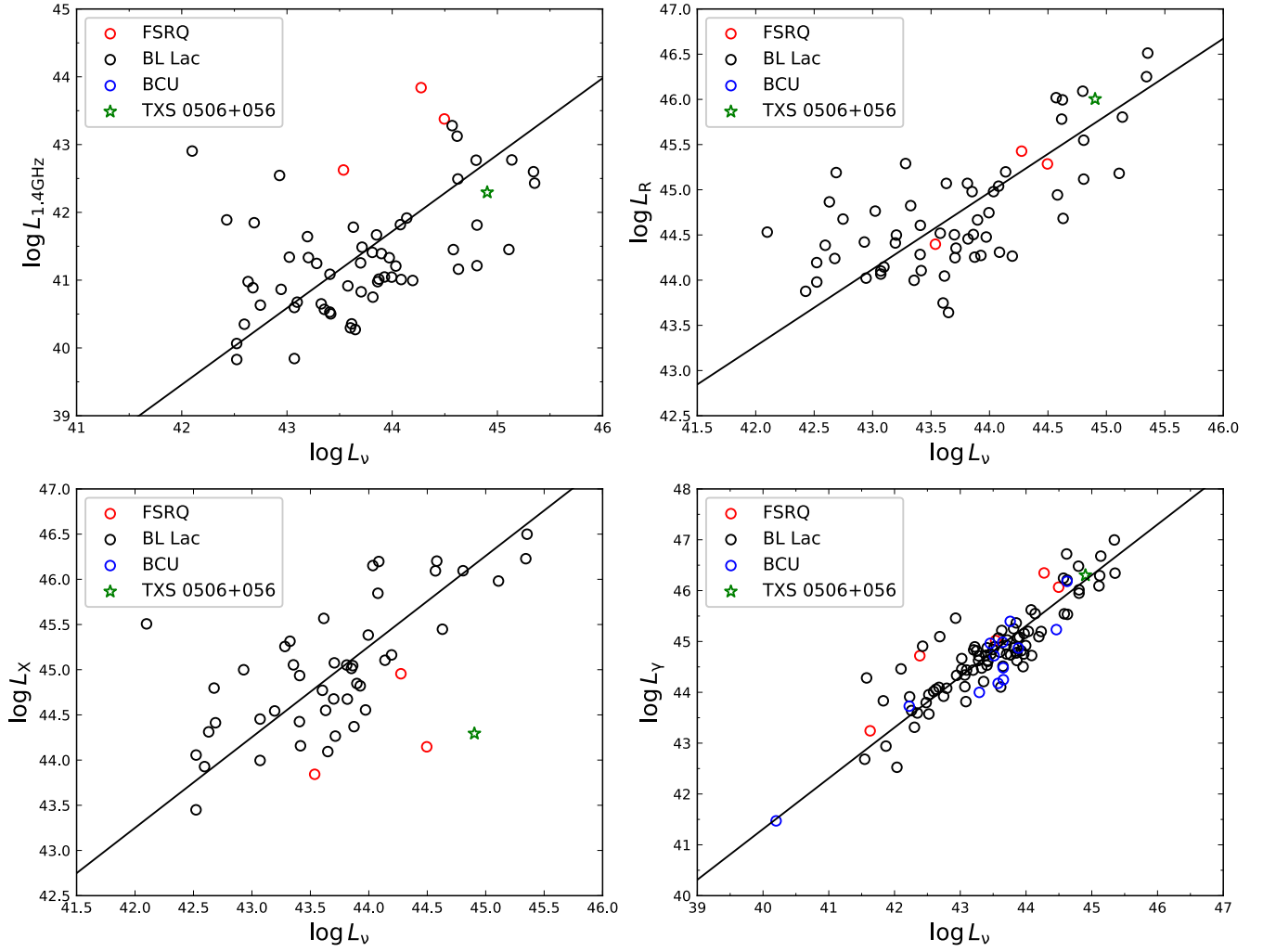


Figure 3. Electromagnetic luminosity as a function of neutrino luminosity. The red circles represent FSRQs, the black circles denote BL Lac objects, and the blue circles indicate BCUs. The green star marks TXS 0506+056, which is excluded from the linear regression, and the solid black line shows the best-fit linear regression.

Table 4
Regression Results for Neutrino Luminosity versus Electromagnetic Luminosity

Correlation (1)	N (2)	β (3)	α (4)	r (5)	p (6)
$\log L_{1.4 \text{ GHz}}$ versus $\log L_\nu$	60	1.13 ± 0.10	-8.00 ± 4.22	-0.13	0.33
$\log L_R$ versus $\log L_\nu$	60	0.85 ± 0.07	7.56 ± 2.85	0.30	0.02
$\log L_X$ versus $\log L_\nu$	48	1.00 ± 0.08	1.12 ± 3.50	0.52	1.8×10^{-4}
$\log L_\gamma$ versus $\log L_\nu$	103	1.00 ± 0.04	1.37 ± 1.89	0.78	9.1×10^{-22}

Note. Column (1): correlation. Column (2): number of sources. Column (3): slope. Column (4): intercept. Column (5): partial Pearson correlation coefficient. Column (6): chance probability.

estimated primarily using emission-line properties, while absorption-line features and bulge luminosity are additionally employed to constrain L_{disk} . The correlations are examined using linear relations of the form $y = \beta x + \alpha$, derived through OLS bisector regression. The regression results are presented in Figure 4 and summarized in Table 5. In addition, a partial-correlation analysis is performed to evaluate the intrinsic correlation between $\log L_\nu$ and $\log L_{\text{disk}}$.

No significant linear correlation is found for either $\log M_{\text{BH}}$ versus $\log L_\nu$ or $\log L_{\text{disk}}$ versus $\log L_\nu$. For $\log M_{\text{BH}}$ and $\log L_\nu$, the Pearson correlation coefficient is $r = 0.06$, with

a chance probability of $p = 0.72$, indicating the absence of a meaningful correlation. The relationship between $\log L_{\text{disk}}$ and $\log L_\nu$ appears to be more complex. After accounting for the redshift dependence inherent in luminosity–luminosity correlations, the partial-correlation analysis yields $r = 0.24$ and $p = 0.16$, suggesting that the correlation is not statistically significant. Nevertheless, a weak positive trend between $\log L_{\text{disk}}$ and $\log L_\nu$ is visually apparent in the right panel of Figure 4, which may hint at an underlying connection that cannot be firmly established with the current sample.

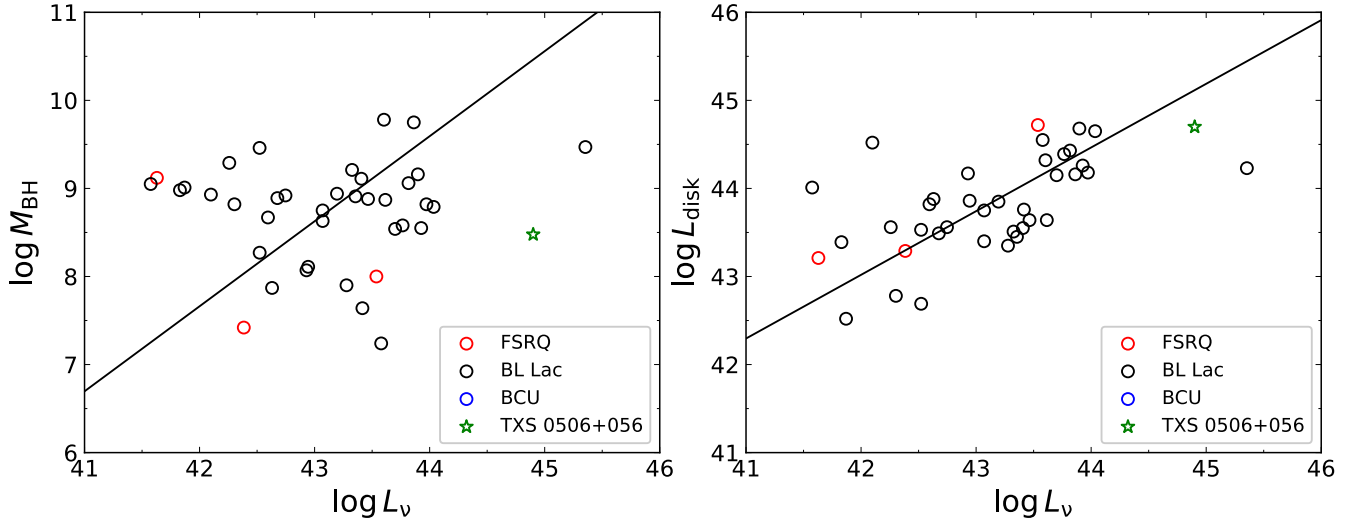


Figure 4. Central engine as a function of neutrino luminosity. The symbols used for the data points and the regression lines are the same as those adopted in Figure 3.

Table 5
Regression Results for Central Engine versus Neutrino Luminosity

Correlation (1)	N (2)	β (3)	α (4)	r (5)	p (6)
$\log M_{\text{BH}}$ versus $\log L_{\nu}$	38	0.97 ± 0.09	-32.89 ± 3.84	0.06	0.72
$\log L_{\text{disk}}$ versus $\log L_{\nu}$	38	0.72 ± 0.09	12.65 ± 3.91	0.24	0.16

Note. Column (1): correlation. Column (2): number of sources. Column (3): slope. Column (4): intercept. Column (5): Pearson correlation coefficient. Column (6): chance probability.

4. Discussion

4.1. The Hadronic Model of This Work

In hadronic models, both electrons and protons are accelerated to relativistic energies, with protons exceeding the threshold for $p\gamma$ photo-pion production on the ambient soft photon field. The lower-energy hump is still dominated by electron synchrotron radiation, whereas the higher-energy hump is mainly produced by proton synchrotron emission, π^0 decay, and electromagnetic cascades initiated by secondary particles (K. Mannheim & P. L. Biermann 1992; F. A. Aharonian 2000; A. Mücke et al. 2003; M. Böttcher et al. 2013). In this work, we employ a hadronic model to reproduce the HE component of our NBCs, using multiwavelength data collected from public databases. Compared to previous hadronic models in the literature (e.g., A. Zech et al. 2017; X. Rodrigues et al. 2024), in which the GeV emission observed by Fermi-LAT is predominantly attributed to proton synchrotron radiation while cascade emission accounts for the radiation beyond the GeV band, our modeling suggests a different scenario. In our case, the proton synchrotron component is shifted to the MeV range for most NBCs, whereas the GeV emission is primarily produced by synchrotron radiation from Bethe–Heitler pairs. The role of Bethe–Heitler pair production in shaping the γ -ray peak has been investigated in detail by D. Karavola & M. Petropoulou (2024), who showed that blazars with the lower-energy hump located at $10^{14-15.5}$ Hz and the higher-energy hump peaking at 10^{22-25} Hz are more likely to have their HE emission dominated by Bethe–Heitler processes. Our sample broadly falls within this regime, supporting such an interpretation. Compared to their results, our modeling is characterized, on average, by a larger emission-region size (R)

and a smaller Doppler factor. In addition, we adopt a fixed proton spectral index of 1, which further distinguishes our parameter setup from previous studies. With this parameter setup, the proton synchrotron bump is generally located in the MeV range and has a higher intensity than the Bethe–Heitler pair bump in the GeV range, as discussed in Section 4.3.

Hadronic models offer several advantages in addressing challenges encountered in one-zone leptonic models, such as extremely rapid variability and the requirement of very high bulk Lorentz factors (F. Aharonian et al. 2007; M. C. Begelman et al. 2008; Z. Ouyang et al. 2025), VHE excess emission (MAGIC Collaboration et al. 2008; M. Böttcher et al. 2013; J.-G. Cheng et al. 2022), and neutrino production in blazars (M. Cerruti et al. 2019; R. Xue et al. 2021). Nevertheless, hadronic models are often challenged by significant deviations from equipartition between the dominant particle component and the magnetic field (e.g., M. Böttcher et al. 2013), as well as by the requirement of super-Eddington jet powers (e.g., J.-G. Cheng et al. 2022). In this work, we find that the power carried by relativistic particles dominates over that of the magnetic field by 1–2 orders of magnitude. We define the equipartition parameter as $\epsilon_{pB} \equiv P_p/P_B$, and an average value of $\log \epsilon_{pB} \simeq 1.67 \pm 1.03$ is found for the NBCs in our sample. The maximum value, $\log \epsilon_{pB} = 5.27$, is found for 4FGL J0509.9–6417, corresponding to a relatively weak magnetic field strength of $\log B = -0.57$. The minimum value, $\log \epsilon_{pB} = -0.36$, is obtained for 4FGL J1719.3+1205, where comparable powers are inferred for the magnetic field ($\log P_B = 46.35$) and protons ($\log P_p = 45.89$). Among the 103 NBCs in our sample, 10 sources exhibit $-0.5 \leq \log \epsilon_{pB} \leq 0.5$, indicating approximate energy equipartition between particles and the magnetic field. For the 39

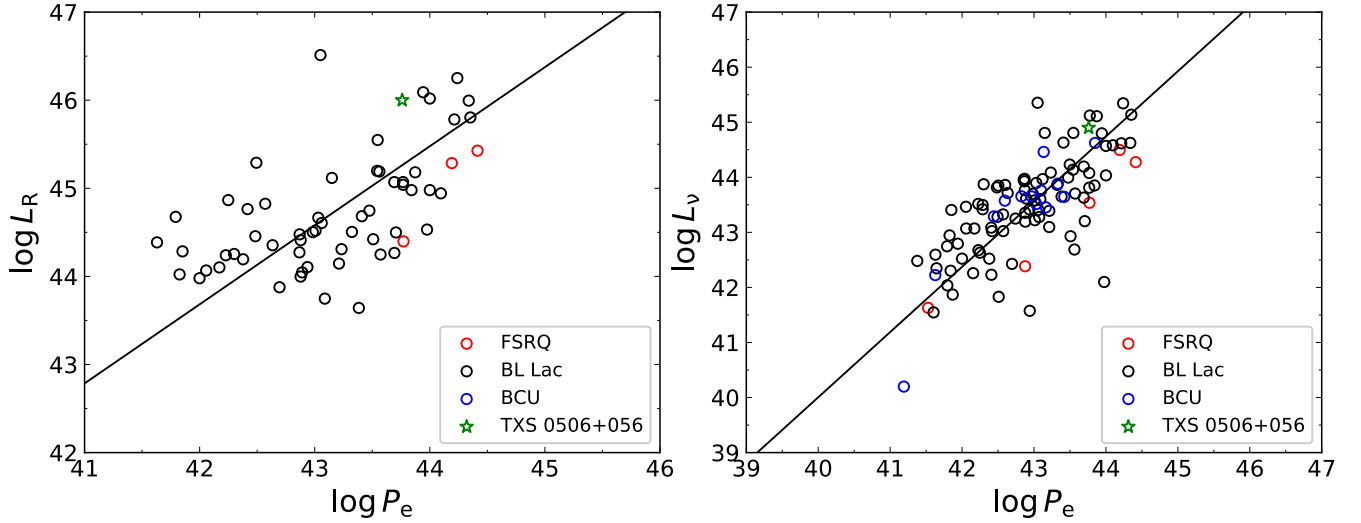


Figure 5. Optical luminosity (left) and neutrino luminosity (right) as a function of the electron luminosity. The symbols used for the data points and the regression lines are the same as those adopted in Figure 3.

Table 6
Linear Regression Results for Electron Power versus Neutrino and Optical Luminosities

Correlation (1)	N (2)	β (3)	α (4)	r (5)	p (6)
$\log L_R$ versus $\log P_e$	60	0.90 ± 0.09	6.00 ± 3.73	0.58	1.3×10^{-6}
$\log L_\nu$ versus $\log P_e$	103	1.18 ± 0.08	-7.37 ± 3.28	0.72	1.1×10^{-17}

Note. Column (1): correlation. Column (2): number of sources. Column (3): slope. Column (4): intercept. Column (5): Pearson correlation coefficient. Column (6): chance probability.

sources with available black hole mass estimates, 14 sources show total jet powers exceeding the Eddington luminosity, $L_{\text{Edd}} = 1.26 \times 10^{38} (M_{\text{BH}}/M_\odot) \text{ erg s}^{-1}$. The P_{tot} to L_{Edd} ratio ranges from a minimum of 0.01 to a maximum of 2.55, suggesting that a significant fraction of the sources operate at or above the Eddington limit. The issue of exceeding the Eddington limit may be reconciled by invoking multizone models. For instance, X. Rodrigues et al. (2026) recently proposed a physically motivated, continuous multizone blazar jet lepto-hadronic model, in which particle acceleration and radiation evolve along the jet, and successfully calculated the SEDs of TXS 0506+056 and PKS 0605–085.

4.2. Correlations between Neutrino Emission and Other Parameters

Among the examined correlations between neutrino luminosity and electromagnetic luminosity, the relationship between $\log L_\nu$ and $\log L_{1.4\text{GHz}}$ yields a chance probability of $p = 0.33$, indicating no statistically significant correlation between neutrino emission and radio-band emission. In contrast, the correlations of $\log L_\nu$ with $\log L_X$ and $\log L_\nu$ with $\log L_\gamma$ are both strong. This behavior is expected if the X-ray and γ -ray emissions originate from hadronic processes. Considering that the X-ray and γ -ray fluxes are independently compiled from BZCAT and represent the average emission levels in these bands, the observed significant correlations imply that a hadronic origin of the X-ray and γ -ray emission in blazars cannot be completely excluded.

An intriguing result among these correlations is the weak-to-moderate correlation between $\log L_\nu$ and $\log L_R$. Within our SED modeling framework, the optical emission is primarily

attributed to synchrotron radiation from relativistic electrons, a purely electromagnetic process that does not produce neutrinos. Nevertheless, several factors may contribute to the observed correlation. First, this trend may arise from sample-selection effects and could be modified or weakened when a larger and more complete sample is considered. If the correlation has a physical origin, it may result from multiple contributing mechanisms. In particular, the optical emission in the R band is composed of radiation from both the jet and the accretion disk, with the jet component generally dominating.

A jet-related origin of this correlation is a natural consequence if the optical emission provides seed photons for photo-pion production, thereby leading to subsequent neutrino emission, provided that protons are accelerated to sufficiently high energies. The correlations between the jet electron power ($\log P_e$) and $\log L_R$, as well as between $\log P_e$ and $\log L_\nu$, are shown in Figure 5, with the corresponding regression results listed in Table 6. These correlations support our proposed interpretation. In the left panel of Figure 5, we find a positive correlation between $\log L_R$ and $\log P_e$, where $P_e \simeq \Gamma^2 L_e$, indicating that the optical emission is indeed dominated by synchrotron radiation from relativistic electrons in the jet. In the right panel of Figure 5, a positive correlation is also observed between $\log L_\nu$ and $\log P_e$, suggesting that synchrotron photons produced by relativistic electrons act as seed photons for photo-pion ($p\gamma$) interactions.

In addition, a disk-related origin of this correlation cannot be excluded. Neutrino emission may be related to the optical emission through the accretion disk if a fraction of the optical R -band emission in blazars originates from the disk. In this case, the optical R -band luminosity can serve as a proxy for the

disk luminosity, such that stronger R -band emission corresponds to a more powerful accretion disk. Under this assumption, the observed correlation between $\log L_R$ and $\log L_\nu$ can be interpreted as a manifestation of an underlying correlation between the disk luminosity, $\log L_{\text{disk}}$, and the neutrino luminosity, $\log L_\nu$. The correlation between $\log L_{\text{disk}}$ and $\log L_\nu$, shown in Figure 4 and quantified in Table 5, suggests the presence of a weak positive trend. If the relatively large p -value is primarily due to the limited sample size (38 sources), and if a positive correlation between these two quantities indeed exists, it may be understood in two possible ways. On the one hand, photons associated with the accretion disk and/or corona—such as X-ray or optical photons—may serve as seed photons for photo-pion interactions (R. Xue et al. 2021; K. Murase 2022). On the other hand, blazar neutrino emission may originate from the inner regions of the accretion disk or the corona (i.e., the AGN core), provided that hadrons are accelerated together with leptons (M. C. Begelman et al. 1990; Y. Inoue et al. 2019). Previous studies have suggested that the AGN core itself could act as a site of HE neutrino production (F. W. Stecker 2013; O. Kalashev et al. 2015), although such emission may be insufficient to account for the IceCube detection associated with TXS 0506+056 (D. F. G. Fiorillo et al. 2025).

4.3. Particle Composition of the Blazar Jet

Leptonic models have long been established as the standard framework for explaining blazar emission, in which low-energy radiation is produced by electron synchrotron emission and HE photons arise from IC scattering. Motivated by the proton-dominated composition of cosmic rays, hadronic models were subsequently proposed, in which relativistic protons in jets interact with ambient photon fields to produce HE γ -rays and neutrinos. Although hadronic models can naturally account for TeV emission and neutrino production, they often require jet powers exceeding the Eddington limit. The detection of HE neutrinos from blazars has provided compelling multimessenger evidence for the presence of relativistic hadrons in blazar jets.

4.3.1. Emission in the MeV Range

In this work, we adopt a hadronic model to construct the broadband SEDs of 103 NBCs and to investigate the physical properties of blazar neutrino emission. To obtain the highest possible neutrino flux, we impose the condition that the HE emission produced by leptonic IC scattering is suppressed to a level 3 orders of magnitude lower than that arising from $p\gamma$ interactions. To achieve this, the electron power is reduced while the magnetic field strength is increased, thereby enhancing synchrotron cooling of electrons. Consequently, the proton energy budget must be increased to reproduce the observed GeV–TeV γ -ray emission. As a result, synchrotron radiation from relativistic protons produces a distinct spectral bump in the broadband SED, located in the MeV energy range. Indeed, for 99 out of the 103 NBCs in our sample, the proton synchrotron peak is found within the 0.1–100 MeV band, with an average proton energy $\log \gamma_p = 7.89 \pm 0.34$. Among these sources, 90 exhibit a significant proton synchrotron peak, with MeV-band proton synchrotron fluxes exceeding the Bethe–Heitler pair-production flux in the GeV band. This distinctive spectral feature and the associated strong MeV emission are

expected to be detectable by next-generation MeV γ -ray missions, such as e-ASTROGAM (A. De Angelis et al. 2018) and AMEGO (J. McEnery et al. 2019). Moreover, the MeV bump can be tightly constrained through joint observations with future MeV γ -ray facilities and next-generation X-ray instruments, e.g., the enhanced X-ray Timing and Polarization telescope (or eXTP; P. Zhou et al. 2025), which will provide broad energy coverage from the soft X-ray band (keV) through hard X-rays and extending into the MeV regime (up to ~ 600 keV). Such multiwavelength observations will enable stringent tests of the hadronic scenario adopted in this work and provide valuable insights into the particle composition of blazar jets.

4.3.2. Neutrino Emission

Defining the neutrino-to- γ -ray intensity ratio as $Y_{\nu\gamma} = L_\nu/L_\gamma$ (P. Padovani et al. 2015), where L_ν is the peak neutrino luminosity and L_γ is the integrated γ -ray luminosity in the 1–100 GeV band, we obtain an average value of $\langle Y_{\nu\gamma} \rangle = 0.074 \pm 0.067$, with a maximum value of 0.32. We find that 14 sources exhibit $Y_{\nu\gamma} > 0.13$, a value previously suggested as an approximate upper limit for blazars based on 7 yr IceCube ultra-high-energy data (M. G. Aartsen et al. 2016). These sources tend to show relatively higher L_ν and lower L_γ compared to the overall NBC sample. In addition, we perform a rough estimate of the contribution of NBCs to the diffuse neutrino background using

$$E_\nu^2 \Phi_\nu = \frac{1}{4\pi} \frac{N_{\text{sample}}}{N_{\text{all}}} \sum_i (\nu_i F_i), \quad (4)$$

where $\nu_i F_i$ represents the neutrino flux of individual sources, $N_{\text{sample}} = 103$ is the sample size in this work, and $N_{\text{all}} = 199$ corresponds to the full NBC sample in Paper I. A comparison between our estimate and the IceCube stacking limit for blazars is shown in Figure 6. The result indicates that our scenario, which explores the maximum neutrino emission capability of blazars, is not excluded by current observational constraints.

Based on our SED results, the peak neutrino fluxes of these NBCs range from 1.55 PeV to 275.42 PeV. This energy range can be covered by current and next-generation neutrino facilities, such as IceCube (M. G. Aartsen et al. 2017), the Cubic Kilometre Neutrino Telescope (KM3NeT; S. Adrián-Martínez et al. 2016), the NEutrino Observatory in the Nanhai (NEON; H. Zhang et al. 2025), IceCube Generation 2 (IceCube-Gen2; M. G. Aartsen et al. 2021), and Tropical Deep-sea Neutrino Telescope (TRIDENT; Z. P. Ye et al. 2023). We compile the 5σ discovery potential fluxes of these neutrino facilities, which vary with decl., and compare them with the peak neutrino fluxes predicted by our model in Figure 7 and list the results in Table 7. We find that three NBCs are potentially detectable by IceCube, 20 by IceCube-Gen2, 22 by KM3NeT, 35 by NEON@1700m, 45 by NEON@3500m, and 62 by TRIDENT. Consequently, some of these sources could be confirmed as neutrino emitters by KM3NeT, NEON, and TRIDENT in the coming years.

Four BL Lac objects, TXS 0506+056 and three of our NBCs, lie above the IceCube discovery potential curve. Within our model, the peak neutrino flux of TXS 0506+056 exceeds the IceCube discovery potential by nearly 1 order of magnitude, consistent with its confirmed neutrino detection. The remaining three NBCs—4FGL J0045.7+1217, 4FGL

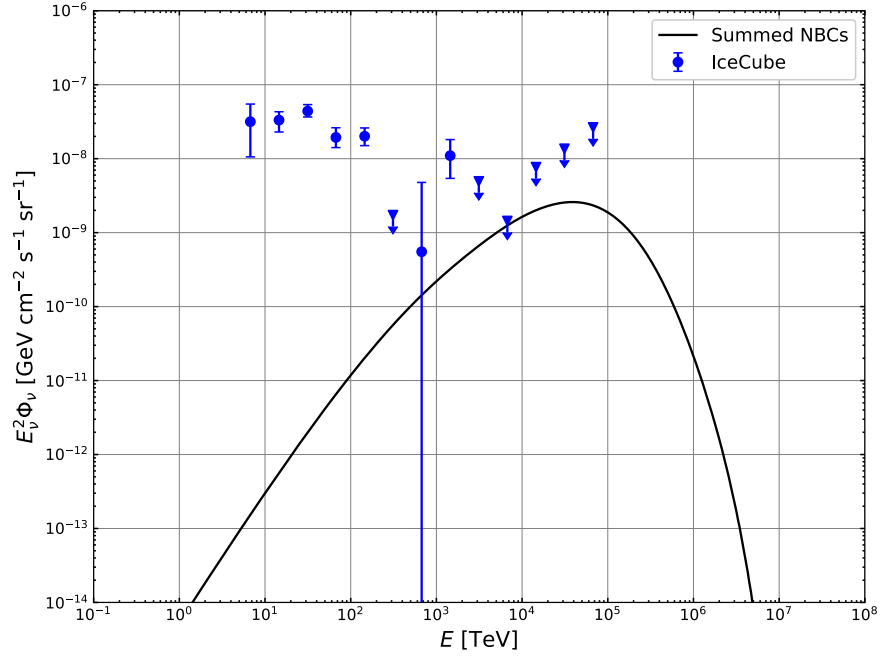


Figure 6. The summed neutrino flux as a function of energy. The blue data points and upper limits represent the best-fit results of the differential flux model for the 2010–2015 (6 yr) IceCube cascade data, while the curve represents the summed neutrino flux from the NBCs.

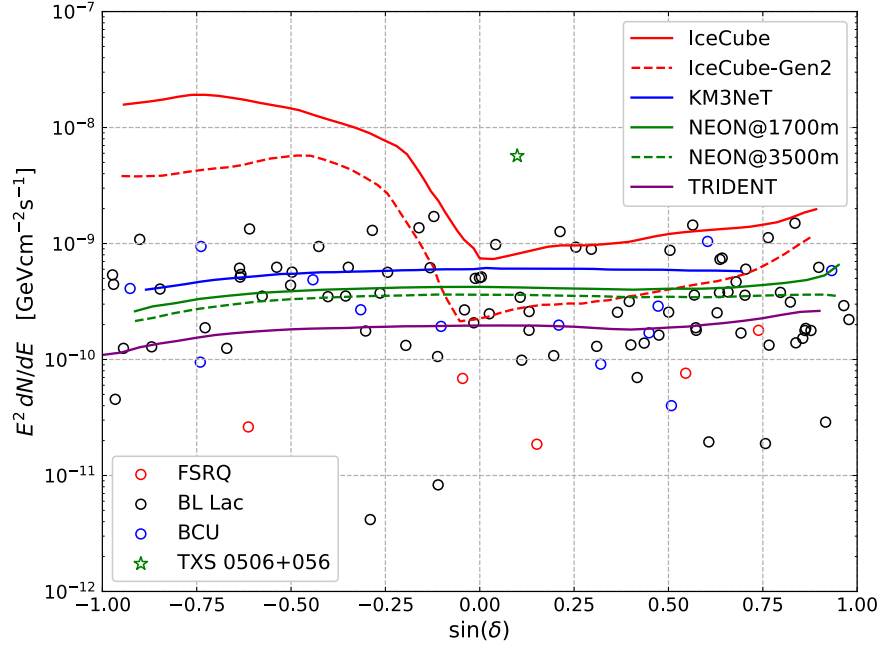


Figure 7. The 5σ discovery potential of our neutrino blazar candidates (NBCs) for current and next-generation neutrino detectors as a function of decl., based on a 10 yr exposure. The solid red curve represents the 5σ discovery potential for IceCube (M. G. Aartsen et al. 2020), the dashed red curve for IceCube-Gen2 (A. Omeliukh et al. 2022), the solid blue curve for KM3NeT (S. Aiello et al. 2019), the solid and dashed green curves for NEON (H. Zhang et al. 2025), and the solid purple curve for TRIDENT (Z. P. Ye et al. 2023). The symbols used for the data points are the same as those adopted in the previous figures.

J0123.1+3421, and 4FGL J0326.2+0225—only marginally exceed the IceCube sensitivity curve, yet they may still be detectable in neutrinos. However, none of these three NBCs has been reported as a neutrino emitter at the time of writing. This discrepancy is expected, as the neutrino fluxes derived in our model represent upper limits, and the actual neutrino emission may be significantly weaker. The SEDs of these four sources are shown in Figure 8; we note that these sources all show strong neutrino emission comparable to their γ -ray flux.

At present, TXS 0506+056 remains the only blazar firmly confirmed as a neutrino emitter in the PeV range. More recently, the Seyfert II galaxy NGC 1068 has also been identified as a neutrino source, with a significance of 4.2σ in the TeV band (IceCube Collaboration et al. 2022). This discovery has prompted discussions on whether HE neutrinos may predominantly originate from the accretion disk corona rather than from relativistic jets in AGNs. Consequently, the identification of additional blazars as neutrino emitters would

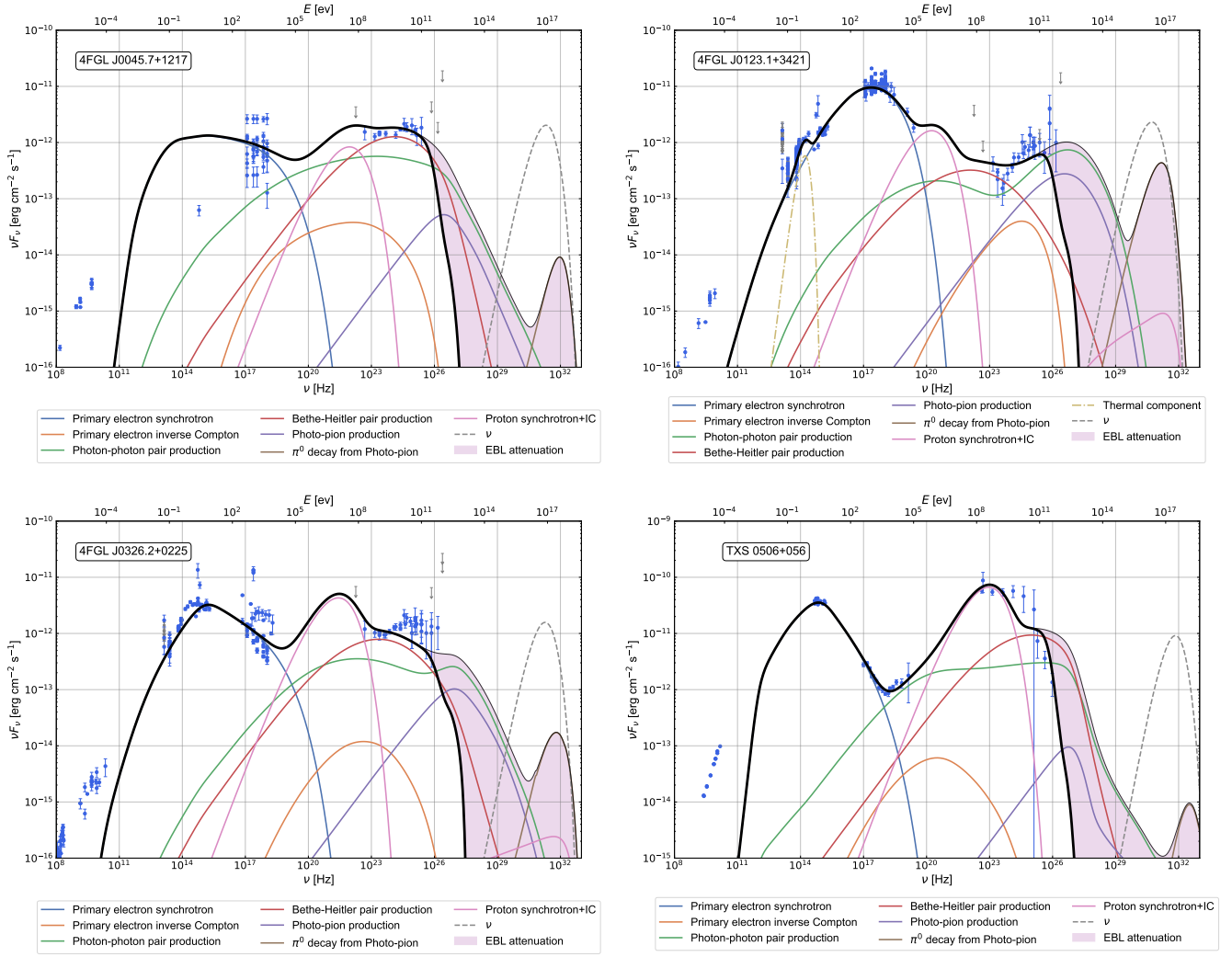


Figure 8. SED modeling plots for TXS 0506+056 and three NBCs. The symbols used for the data points and the curves are the same as those adopted in Figure 1.

Table 7
Potential Detectability of Our NBCs by Neutrino Facilities

4FGL Name (1)	IceCube (2)	IceCube-Gen2 (3)	KM3NeT (4)	NEON@1700m (5)	NEON@3500m (6)	TRIDENT (7)
J0006.3–0620	...	...	...	...	...	...
J0029.0–7044	...	...	...	...	...	Y
J0045.7+1217	Y	Y	Y	Y	Y	Y
J0121.8–3916	...	...	Y	Y	Y	Y
J0123.1+3421	Y	Y	Y	Y	Y	Y

Note. Column (1): 4FGL name. Column (2): potential detectability by IceCube, where “Y” denotes “Yes.” Column (3): potential detectability by IceCube-Gen2. Column (4): potential detectability by KM3NeT. Column (5): potential detectability by NEON@1700m. Column (6): potential detectability by NEON@3500m. Column (7): potential detectability by TRIDENT. Note, only five items are displayed.

(This table is available in its entirety in machine-readable form in the [online article](#).)

provide a crucial smoking-gun signature for the presence of relativistic hadrons in blazar jets and neutrino emission regions in AGNs.

5. Conclusion

In the present work, we extend the study of neutrino blazars following our previous investigation (Paper I). We model the

broadband SEDs constructed from archival multiwavelength data for 103 sources—comprising 85 BL Lacs, five FSRQs, and 13 BCUs—within a hadronic framework. Our main conclusions can be summarized as follows. (1) From the broadband SED modeling, we constrain nine free parameters that characterize the physical properties of the emission region and the energy distributions of relativistic particles in the jets of these 103 NBCs. (2) Using a partial-correlation analysis, we

investigate the relationship between neutrino luminosity and multiwavelength electromagnetic emission (radio, optical R band, X-ray, and γ -ray). The regression results reveal positive correlations between $\log L_\nu$ and $\log L_R$, $\log L_X$, and $\log L_\gamma$. (3) The observed correlation between $\log L_R$ and $\log L_\nu$ provides potential clues to the neutrino production site. The neutrino production site may be located either within the jet or associated with the disk–corona system, although it cannot be further constrained with the current analysis. (4) Our results highlight the MeV band as a key observational window for distinguishing between leptonic and hadronic emission scenarios, as well as for constraining the particle composition of blazar jets. (5) Based on the maximum neutrino fluxes predicted by our hadronic model, three NBCs are expected to be detectable by IceCube, while up to 22 NBCs may be within the detection reach of KM3NeT in the near future.

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